

Calc 2 – Power Series

Worksheet A

1. Find the radius and interval of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
2. Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.
3. Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n! x^n$.
4. Differentiate the series $\sum x^n$ ($|x| < 1$) to give a closed form for $\sum_{n=1}^{\infty} nx^{n-1}$.
5. Integrate $\frac{1}{1-x} = \sum x^n$ to obtain a power series for $-\ln(1-x)$ and state its radius.

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Worksheet B

1. Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.
2. Find the radius and interval of convergence of $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$.
3. Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$.
4. Use $\frac{1}{1-x} = \sum x^n$ to write $\frac{1}{1+x^2}$ as a power series and give its radius.
5. Substitute into the series for e^x to get a Maclaurin series for e^{2x} .

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Answer Key (Worksheets A & B)

Worksheet A.

1. Ratio: $\frac{|x|^{n+1}/(n+1)!}{|x|^n/n!} = \frac{|x|}{n+1} \rightarrow 0$ for all x . $R = \infty$, interval $\boxed{(-\infty, \infty)}$.
2. Ratio: $|x| \cdot \frac{n}{n+1} \rightarrow |x|$. $R = 1$. Endpoints: $x = 1$: $\sum 1/n$ diverges; $x = -1$: alternating harmonic converges. Interval $\boxed{[-1, 1)}$.
3. Ratio: $\frac{(n+1)! \cdot |x|}{n!} = (n+1)|x| \rightarrow \infty$ unless $x = 0$. $R = 0$, interval $\boxed{\{0\}}$.
4. $\frac{d}{dx} \sum_{n=0}^{\infty} x^n = \sum_{n=1}^{\infty} nx^{n-1}$ and $\frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}$. So $\boxed{\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}, |x| < 1}$.
5. $\int \sum x^n dx = \sum \frac{x^{n+1}}{n+1}$. So $-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$, $R = 1$ (interval $[-1, 1)$).

Worksheet B.

1. Ratio: $\frac{|x-2|}{3} \cdot \frac{n}{n+1} \rightarrow \frac{|x-2|}{3}$. $R = 3$. Endpoints $x = 5$: $\sum 1/n$ diverges; $x = -1$: alternating harmonic converges. Interval $\boxed{[-1, 5]}$.
2. Ratio: $\frac{x^2}{(2n+1)(2n+2)} \rightarrow 0$ for all x . $R = \infty$, interval $\boxed{\mathbb{R}}$. (Series equals $\cos x$.)
3. Root test: $\sqrt[n]{|x|^n/n^n} = |x|/n \rightarrow 0$ for all x . $\boxed{R = \infty}$.
4. Substitute $x \rightarrow -x^2$: $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$. Radius $\boxed{R = 1}$.
5. $e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!} = \sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$, all x .